Mallable World Model

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Abstract

Model-based reinforcement learning agents typically rely on pretrained world models that remain static during deployment. In open-ended settings, however, environmental shifts—such as actuator wear or domain gaps—cause these frozen dynamics to diverge from reality, leading to prediction errors and control degradation. While gradient-based online adaptation can recover competence, it often incurs prohibitive computational costs and risks catastrophic forgetting. We introduce **Malleable World Models (MWM)**, a test-time adaptation framework that decomposes adaptation across three distinct timescales. MWM augments a frozen pretrained world model with a fast, backpropagation-free plastic prior residual that corrects one-step latent dynamics via a local Hebbian update. This fast store is paired with a slow gradient-based actor adaptation pathway and an optional world-model consolidation mechanism that systematically transfers compensatory structure into the pretrained weights. We evaluate MWM on proprioceptive locomotion tasks under controlled actuator gain shifts. Our results demonstrate that fast plasticity effectively detects shifts and provides a stable scaffold for slow policy learning. We further show that our consolidation framework enables the mechanical transfer of adaptation from the fast store to structural weights, providing a robust, multi-timescale architecture for persistent test-time adaptation.

1 Introduction

World models enable model-based agents to plan and act by predicting latent dynamics from past experience [**?** **?**]. However, these models are typically deployed with *frozen* parameters. In open-ended environments—where agents encounter actuator wear, payload variations, or simulation-to-real gaps—the transition kernel internalized during pretraining often diverges from the actual deployment environment. Consequently, prediction errors escalate, imagined rollouts become misleading, and control performance severely degrades. *Test-time adaptation* (TTA) aims to recover agent competence under such shifts without resorting to expensive offline retraining.

The dominant paradigm for TTA relies on gradient-based online learning: collecting post-shift trajectories, backpropagating a surrogate loss, and directly updating network weights [**?** **?**]. While expressive, this *slow* adaptation pathway incurs significant operational costs in interactive control, including batched updates, autograd computation over long horizons, and extensive hyperparameter tuning to mitigate catastrophic forgetting. When environmental shifts occur abruptly, the agent inevitably exhibits suboptimal behavior for numerous steps before the initial weight updates can take effect.

Biological systems circumvent this limitation by separating a fast, local, error-driven memory store from slower structural learning [**?**]. Rather than seeking global optimality, this fast store provides coarse, immediate corrections to sustain functional behavior during transient phases. Aligning with this architecture, we argue that test-time adaptation in world models should manifest not merely as an immediate motor reflex, but as a rapid recalibration of *what the agent expects to happen next*.

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We introduce **Malleable World Models (MWM)**, a framework that augments frozen pretrained agents with a **fast plastic prior residual** P_t. The residual is updated at every environment step via a backpropagation-free Hebbian rule, operating independently of the pretraining weights. This fast pathway is optionally paired with a **slow** pathway that updates the actor and world-model weights from a replay buffer. We evaluate MWM on the proprioceptive `walker-walk` task under sudden actuator gain shifts ($1.0 \rightarrow 0.7 \rightarrow 1.0$). Section 3 details the architectural design, and Section 4 characterizes its shift detection, adaptation latency, and consolidation behavior.

Contributions.

1. We design a backpropagation-free plastic prior on RSSM logits featuring a norm-capped fast state matrix P_t.
2. We propose a tri-timescale test-time architecture comprising fast step-level plasticity (P_t), slow interval-based actor fine-tuning, and long-term world-model consolidation with a coupled flush mechanism.
3. We empirically demonstrate that under gain shifts, the fast pathway provides rapid trace growth and prior-error correction. This fast scaffold materially enhances slow actor learning, while pilot results validate the systematic drainage of P_t during consolidation.

2 Related Work

MWM operates at the intersection of model-based reinforcement learning, test-time adaptation, and dual-timescale memory systems.

2.1 World models and model-based RL

Learned dynamics models achieve sample-efficient control by facilitating trajectory imagination within a latent space [**? ? ?**]. Dreamer-style architectures pair a recurrent state-space model (RSSM) with an actor–critic network optimized via these imagined rollouts [**? ?**]. R2-Dreamer and associated redundancy-reduced variants streamline representation learning while maintaining latent imagination capabilities [**?**]. Because latent-space planners [**? ?**] depend heavily on transition accuracy, environmental drift introduces severe biases into imagined rewards and values, even if the policy architecture remains functional. Unlike standard agents that keep dynamics fixed post-pretraining, MWM introduces a separate fast state P_t over prior logits, keeping the bulk of the parameters θ frozen while optionally fine-tuning a small subset θ_{wm} under a consolidation schedule.

2.2 Test-time adaptation and online fine-tuning

Meta-reinforcement learning traditionally optimizes initializations or update rules to enable rapid adaptation via few gradient steps at deployment [**? ?**]. However, these methods still depend on online backpropagation through policy or model parameters, requiring carefully curated task distributions and inner-loop learning rates. In supervised settings, test-time adaptation shifts batch-normalization statistics or encoder weights using unlabeled test streams [**? ?**]. Analogously, online RL methods continue policy or model training using post-shift trajectories collected in deployment [**? ? ?**]. While frameworks like RL^2 embed adaptation directly into the recurrent hidden state [**?**], they necessitate exhaustive meta-training across explicit episodic variations.

A common denominator across these approaches is their reliance on a global adaptation mechanism where weight updates propagate through backpropagation, risking catastrophic forgetting [**?**]. MWM addresses this by introducing a local store (P_t) that updates at every step from prediction errors without autograd. Our experiments demonstrate that this fast pathway acts as a complementary scaffold that accelerates and stabilizes slow gradient-based actor learning.

2.3 Fast memory, context, and dual-process learning

Several architectures partition transient context from persistent parameter stores. Hypernetworks and fast weights modulate connectivity dynamically based on input or episodic history [**? ? ?**], while key–value episodic memories retrieve past transitions for few-shot control [**? ?**]. In sequence

Table 1: Positioning of MWM relative to adjacent approaches.

Approach	Fast store	Slow update	Targets WM prior
Meta-RL / MAML [**?**]	–	grad, all weights	optional
Online fine-tuning [**? ?**]	–	grad, batched	sometimes
RL^2 / episodic memory [**? ?**]	hidden / KV	grad at meta-train	indirect
LoRA / adapters [**? ?**]	–	grad, pretrained	any layer
MWM (ours)	P_t **Hebbian**	**actor /** θ_{wm}	**prior logits**

modeling, in-context learning implicitly adjusts agent behaviors without parameter modifications [**?**]. Complementary Learning Systems (CLS) theory in cognitive science suggests a similar partition: a fast hippocampal store for rapid encoding and a slow neocortical store for gradual integration [**? ?**].

MWM instantiates this CLS framework directly within a world model. P_t serves as the capacity-limited fast trace driven by local errors, while the actor and θ_{wm} act as slow stores optimized via replay. The coupled flush mechanism (Section **??**) serves as an algorithmic analogue for transferring hippocampal traces to cortical weights. Crucially, P_t is explicitly bound to the RSSM prior prediction error, rendering it an interpretable dynamics compensator rather than an arbitrary contextual embedding.

2.4 Hebbian plasticity and non-gradient updates

Hebbian and covariance-based rules update synaptic strengths using local pre- and post-synaptic activity, bypassing global loss minimization [**? ?**]. Outer-product updates of the form $\Delta W \propto e \otimes x$ are widely leveraged in linearized adaptation and fast-weight recurrent networks [**? ?**]. MWM's write rule (Eq. (6)) belongs to this family, utilizing the signed error e_t of prior logits and a latent state-action vector x_t to compute a rank-one increment, bounded by Frobenius-norm projection for stability.

This mechanism fundamentally differs from parameter-efficient fine-tuning methods like LoRA [**?**] or adapters [**?**]. Those approaches introduce trainable parameters that require gradient updates during fine-tuning. In contrast, P_t contains no pretrained weights, requires zero test-time backpropagation, and can be optionally reset at episode boundaries.

2.5 Adapting world models after deployment

Existing frameworks adapt dynamics online for model predictive control [**? ?**], fine-tune latent models under domain shifts [**?**], or retrain decoders for sim-to-real transfer [**?**]. While Dreamer-style agents can theoretically continue world-model training via replay, unconstrained joint updates of dynamics and policies often induce instability under non-stationarity, degrading imagination quality before control performance can recover. MWM structurally constrains the update rates and domains: fast prior corrections are restricted to P_t, slow policy gradients are updated via frozen-world-model imagination (Section **??**), and slow structural updates to θ_{wm} are synchronized systematically. Our ablations show that separating one-step belief corrections from multi-step planning dynamics is essential for preserving policy stability.

2.6 Positioning summary

Table 1 contextualizes MWM against adjacent methodologies. MWM is distinctively framed as a test-time protocol for frozen pretrained backbones, augmenting RSSM priors with a norm-bounded Hebbian residual paired with a tri-timescale consolidation pipeline.

3 Method

We build upon a pretrained model-based agent equipped with a recurrent state-space model (RSSM) [**?**]. At test time, pretrained parameters are frozen for standard inference; downstream learning dynamically modulates a separate fast plastic state P_t, the actor network, or a targeted subset of world-model weights according to distinct temporal schedules.

3.1 Preliminaries: frozen pretrained agent

Let θ denote the parameters of the encoder f_θ, RSSM (h_θ, g_θ), decoder, and policy π_θ, pretrained offline on a nominal task $\mathcal{T}_0$. During deployment, we maintain frozen copies $(f, h, g, \pi) \leftarrow (f_\theta, h_\theta, g_\theta, \pi_\theta)$ for action selection and latent rollouts, unless explicitly modified by a slow optimization pathway.

For any observation o_t, the encoder yields an embedding $e_t = f(o_t)$. The RSSM tracks a stochastic latent state s_t and a deterministic hidden state d_t. The observation step generates a posterior logit ℓ_t^{post} and updates the tuple (s_t, d_t). The corresponding one-step prior over the subsequent stochastic state is parameterized as:

$$\ell_t^{\text{prior}} = g(d_t) \in \mathbb{R}^K, \tag{1}$$

where K represents the flattened stochastic logit dimension. For categorical latents, $K = N_{\text{categ}} \times N_{\text{classes}}$. The policy samples actions via $a_t \sim \pi(\cdot \mid \text{feat}(s_t, d_t))$.

3.2 Environment shift: actuator gain

To isolate the adaptation dynamics, physical simulator parameters (e.g., links, joint friction) are held constant. Environmental shift is introduced via an action-gain wrapper that scales the executed torques:

$$\tilde{a}_t = \gamma_t \, a_t, \qquad \gamma_t \in (0, 1], \tag{2}$$

where a_t is the target action generated by the agent and $\tilde{a}_t$ is the actual torque applied to the environment. This setup encapsulates actuator degradation or uncalibrated control gains without altering the observation space. Our experimental protocol operates in three distinct sequential phases: $\gamma = 1.0$ (in-distribution), $\gamma = 0.7$ (environmental shift), and $\gamma = 1.0$ (recovery).

3.3 Fast pathway: plastic prior-logit residual

State and readout. MWM modulates the frozen prior logits by introducing a test-time matrix $P_t \in \mathbb{R}^{K \times d}$ per parallel environment instance, requiring no extra pretrained parameters. The plastic input features are constructed as:

$$x_t = \left[\text{flat}(s_t);\ d_{t-1};\ \hat{a}_{t-1}\right] \in \mathbb{R}^d, \quad d = K + |d| + |a|, \tag{3}$$

where $\hat{a}_{t-1}$ is the historical action vector, element-wise normalized by $\max(1, |a_{t-1,i}|)$ to preserve numerical stability. The corrected prior logit is expressed as:

$$\ell_t^{\text{prior,pl}} = \ell_t^{\text{prior}} + P_t x_t. \tag{4}$$

During environment interaction, Equation (4) shapes the latent inference trajectory exclusively through the prior structure; the policy continues to sample from states derived via the standard frozen RSSM core.

Signed error. Following the observation update that yields ℓ_t^{post}, we define the clipped signed prediction error as:

$$e_t = \text{clip}\left(\ell_t^{\text{post}} - \ell_t^{\text{prior,pl}}, \rho\right) \in \mathbb{R}^K, \tag{5}$$

where ρ denotes the clipping threshold (default $\rho = 0.5$). We track the Mean Squared Error, $\text{MSE}_t = \|e_t\|_2^2 / K$, as an explicit diagnostic metric for shift detection.

Hebbian write rule. The plastic matrix P_t is updated at every environment step within a `torch.no_grad()` context, completely avoiding backpropagation through the backbone:

$$\Delta P_t = \eta \, g_t \, e_t x_t^\top - \lambda_t^{\text{eff}} P_t, \tag{6}$$

$$P_{t+1} = \text{Proj}_{\|\cdot\|_F \le \mu}\left(P_t + \Delta P_t\right), \tag{7}$$

where η is the learning rate, μ is the Frobenius-norm boundary constraint, and $g_t \in [0, 1]$ is the surprise gating coefficient.

Effective decay and consolidation flush. The system maintains a base decay rate λ_0. The effective decay rate scales as:

$$\lambda_t^{\text{eff}} = \lambda_0 + k_{\text{flush}}\,\sigma\left(-\frac{\|e_t\|_2}{\tau}\right) + b_t, \tag{8}$$

where $\sigma(\cdot)$ is the sigmoid function, τ modulates sensitivity, and $b_t \geq 0$ represents a coupled flush boost that scales upward as the world-model loss improves.

3.4 Online inference loop

At each time step, the agent executes:

1. Compute embedding $e_t = f(o_t)$ and execute the frozen RSSM core to obtain $(\ell_t^{\text{post}}, s_t, d_t)$.
2. Construct x_t, evaluate the plastic residual $P_t x_t$, and output the corrected prior $\ell_t^{\text{prior,pl}}$.
3. Compute e_t and update the fast store P_t via Eqs. (6)–(7).
4. Sample $a_t \sim \pi_\theta(\cdot \mid \text{feat}(s_t, d_t))$ using active parameters.

3.5 Slow pathway: actor adaptation & consolidation

During the shift phase ($\mathcal{B}$), transitions are stored in a replay buffer $\mathcal{D}$. The agent periodically fine-tunes the actor using Dreamer-style imagination [**?**]. Actor adaptation occurs every Δ_{adapt} steps, executing E gradient steps with learning rate α_{actor}.

For consolidation (Table 2), we fine-tune structural world-model weights $\theta_{\text{wm}} \subset \theta$ every Δ_{slow} steps. This updates the encoder and RSSM while systematically increasing the plastic decay rate to flush P_t into the parametric weights θ_{wm}.

Table 2: Test-time conditions for consolidation experiments.

Condition	Online P_t	Actor adapt	Slow θ_{wm}	Flush mode
A (baseline)	✓	✓	–	off
B (slow WM only)	✓	✓	✓	off
C (consolidate)	✓	✓	✓	coupled
D (trap)	✓	–	–	error_only

4 Experiments

We evaluate MWM on the proprioceptive DeepMind Control `walker-walk` task utilizing a frozen pretraining checkpoint (R2-Dreamer, ∼510K environment steps). The operational parameters follow the defaults established in Section 3. We report performance metrics aggregated across 10 distinct random seeds, utilizing paired per-seed rollouts to eliminate environment generation biases.

4.1 Shift detection and fast-store dynamics (Q1–Q2)

Upon encountering the environmental shift ($\gamma = 0.7$), the prior prediction error MSE_t increases abruptly. Concurrently, the fast store trace norm $\|P_t\|_F$ rapidly expands from baseline to its maximum norm boundary ($\mu = 2$) within tens of interaction steps. This profile confirms the execution of the shift detection mechanism (Q1).

Table 3 details the operational metrics under a plastic-only regime compared against the fully frozen baseline during Phase $\mathcal{B}$. The fast plastic pathway successfully minimizes terminal prior MSE relative to the frozen baseline across $6/10$ evaluated seeds. While the magnitude of the MSE reduction exhibits seed-level variance, the fast pathway consistently dampens the immediate reward drop under shift, yielding a superior $\mathcal{B}/\mathcal{A}_1$ return ratio (0.84) compared to the frozen model (0.78), verifying partial latent alignment capabilities (Q2).

A comprehensive ablation over gating strategies demonstrates that while `surprise_only` ($\eta = 0.1$) achieves the highest raw prior-MSE win rate ($8/10$), the `always_on` configuration used across our primary benchmarks remains highly competitive, securing a $7/10$ win rate.

Table 3: Plastic-only vs. frozen inference (n=10 seeds, Phase $\mathcal{B}$). Δr is plastic minus frozen mean return; Δ_{MSE} is frozen minus plastic prior MSE (last 20), so positive favors plastic.

Metric	Value
Mean Δr (plastic $-$ frozen)	-0.009 ± 0.21
Seeds with $\Delta r > 0$	$4/10$
Seeds with $\Delta_{\text{MSE}} > 0$	$6/10$
Mean $\mathcal{B}/\mathcal{A}_1$ return ratio (plastic)	0.84
Mean $\mathcal{B}/\mathcal{A}_1$ return ratio (frozen)	0.78

Table 4: Online P_t during actor adapt (Phase $\mathcal{B}$, n=10). B: plastic on; C: plastic off.

	B (online P_t)	C (no P_t)
Mean return	1.386	1.318
Mean Δr (B $-$ C)	$+0.068 \pm 0.10$	
Seeds with $\Delta r > 0$	$7/10$	

4.2 Control performance and actor adaptation scaffolding (Q3–Q5)

Operating the plastic pathway in isolation does not yield a reliable control advantage; the average return during Phase $\mathcal{B}$ matches the frozen baseline (Table 3), highlighting that fast plasticity alone is insufficient for reliable task control (Q3). Upon returning to nominal parameters ($\mathcal{A}_2$), both configurations smoothly recover their baseline performance profiles without highlighting lingering negative transfer.

Crucially, the fast pathway provides substantial benefits when combined with the slow adaptation loop. Table 4 contrasts actor fine-tuning executed with active online plasticity (Condition B) against fine-tuning performed over a completely frozen inference model (Condition C).

This paired comparison demonstrates that online plasticity successfully scaffolds slow actor adaptation (Q4), securing clear reward performance gains across 70% of evaluated seeds. We attribute this to distributional scaffolding: maintaining P_t during online exploration structures the transition history logged in $\mathcal{D}$, providing a more coherent data manifold for policy optimization.

A blation over imagination configurations reveals that deploying the plastic residual *inside* the multi-step imagination loop consistently degrades policy optimization quality (yielding an average return drop of $\Delta r = -0.13$). As detailed in Table 5, the optimal configuration combines an active online plastic interaction loop with a frozen multi-step imagination model under the `cfg0` cadence (Q5).

Table 5: Actor-adapt configurations (Phase $\mathcal{B}$ mean return, n=10 unless noted).

Config	$(\Delta_{\text{adapt}}, E, \alpha_{\text{actor}})$	Mean $r_{\mathcal{B}}$	Max $r_{\mathcal{B}}$	Δ vs. plastic-only
Plastic-only act	–	1.406	–	–
B_no_adapt	$(50, 4, 1)$ w/o actor grad	1.434	1.594	$+0.027$
cfg0_default	$(50, 4, 1)$	$\mathbf{1.456 \pm 0.14}$	$\mathbf{1.668}$	$\mathbf{+0.050}$
cfg1_conservative	$(100, 2, 0.2)$	1.404 ± 0.11	1.529	-0.003
cfg2_diversity (n=3)	$(200, 4, 0.5)$	1.441	1.501	-0.035
cfg3_micro (n=3)	$(25, 1, 0.1)$	1.416	1.506	-0.060

4.3 Consolidation pilot (Q6)

We evaluate the multi-timescale consolidation pipeline across Conditions A–D using a 3-seed pilot study. As shown in Table 6, Condition C successfully drains the fast store, driving $\|P_t\|_F$ down from its saturation limit to an average of 0.51 by the end of Phase $\mathcal{B}$. This verifies the structural functionality of the coupled flush equations.

However, this mechanical clearance does not immediately produce higher task returns; Condition B (slow updates without flushing) preserves the highest overall reward profile. The dynamics KL

Table 6: Consolidation pilot (n=3, Phase $\mathcal{B}$).

Cond.	Mean return	$\|P_t\|_F$ (last 20)	Prior MSE (last 20)
A baseline	1.357 ± 0.22	2.00	5.91
B slow θ only	$\mathbf{1.409 \pm 0.16}$	2.00	5.03
C consolidate	1.359 ± 0.05	**0.51**	6.06
D flush trap	1.408 ± 0.13	2.00	4.71

loss exhibits non-monotonic oscillations during consolidation updates, suggesting that the permanent parameter weights θ_{wm} did not completely stabilize the shift statistics before the fast store was cleared.

5 Discussion

Our empirical findings outline an asymmetric profile: fast plasticity functions effectively as an online world-model compensator and a distributional scaffold for policy optimization, yet it does not independently drive task-return recovery.

This asymmetry aligns with our structural design. Because MWM's fast pathway modifies the prior logit space rather than directly tuning motor parameters, it re-aligns latent expectations with post-shift observations without rewriting the action selection policy π_θ. If the policy is left unaltered, deploying an isolated plastic residual under major gain shifts can generate imprecise logit corrections that degrade task returns. Thus, the biological parallel to an immediate system reflex describes a mechanism for *belief realignment* rather than direct behavioral adaptation.

The prominent control advantage verified in our benchmarks stems from the interaction between the fast pathway and slow policy adaptation. By maintaining active online plasticity during deployment under shift, the agent stabilizes the data distribution stored within its replay buffer. The slow policy learning loop benefits from this localized calibration. Crucially, our architectural choice to restrict the plastic residual from the multi-step imagination loop prevents the accumulation of roll-out variances, ensuring stable policy gradient trajectories.

The consolidation experiments validate the drainage of P_t, though this mechanical transfer did not translate into immediate reward gains during our pilot runs. We hypothesize three primary factors contributing to this trend. First, a premature flush can trigger rapid drainage before θ_{wm} fully internalizes the shift dynamics, creating a temporary belief gap. Second, the conservative optimization schedule ($\alpha_{\mathrm{wm}} = 0.02$) provides limited updates over the parameter-heavy backbone within the short deployment window. Third, keeping reward and value heads frozen during world-model fine-tuning may introduce structural optimization mismatches.

6 Conclusion

We presented **Malleable World Models (MWM)**, a multi-timescale test-time adaptation protocol for model-based agents. The architecture successfully coordinates three distinct temporal loops: a step-level Hebbian prior logit residual that functions without backpropagation, an interval-based actor fine-tuning pathway, and a long-term world-model parameter consolidation loop.

Our empirical evaluations on actuator gain shifts confirm that while fast plasticity does not independently optimize control returns, it functions as a critical online scaffold that stabilizes the exploration data manifold and significantly enhances gradient-based actor adaptation. Furthermore, we verified that the coupled flush framework successfully clears the fast memory store as structural parameter weights adapt. Refining this multi-timescale interaction to guarantee stable, long-term parametric transfer remains a valuable path for future research in model-based test-time adaptation.

References